

Shahzaib Naseem

AI / ML Engineer | Python · TensorFlow · Deep Learning · NLP · REST APIs
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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer pursuing a B.Sc. in Computer Science (AI specialization) at the University of Education, Lahore. Proficient in building end-to-end machine learning pipelines, designing deep learning architectures (LSTM, Bidirectional LSTM, Attention Mechanism), and deploying models via Flask REST APIs. Experienced in NLP, feature engineering, and integrating AI solutions into full-stack MERN applications. Strong foundation in Python, TensorFlow, Keras, and Scikit-learn with a focus on production-ready, scalable AI systems.

TECHNICAL SKILLS

AI / ML: Python, TensorFlow, Keras, PyTorch, Scikit-learn, LSTM, Bidirectional LSTM, Attention Mechanism, NLP, LLMs, Hugging Face

Data Science: NumPy, Pandas, Matplotlib, Seaborn, MFCC Feature Extraction, Data Augmentation, Hyperparameter Tuning, GridSearchCV

Web / MERN: React.js, Node.js, Express.js, MongoDB, REST APIs, JWT Authentication, HTML5, CSS3, Flask

Tools & Languages: Git, GitHub, Jupyter Notebook, VS Code, Postman | Python, JavaScript, SQL

PROJECTS

Speech Emotion Recognition System | Python · TensorFlow · LSTM · Flask · REST API

- Engineered an end-to-end emotion classification pipeline using MFCC feature extraction on raw audio data, achieving multi-class classification across 6 emotion categories.
- Designed a Bidirectional LSTM + Attention Mechanism architecture; applied data normalization and augmentation to reduce overfitting and improve generalization.
- Deployed via Flask REST API enabling real-time inference from a React.js frontend with sub-second latency; automated full ML workflow reducing manual steps by ~60%.

AI-Powered NLP Chatbot | Python · TensorFlow/Keras · Node.js · React.js · REST API

- Built an intent-recognition chatbot using custom tokenization and a multi-class classification model (TensorFlow/Keras), classifying 30+ intents with high accuracy.
- Developed Node.js/Express.js backend with JWT-secured REST API integrated with a React.js frontend, automating responses for 30+ intent categories.
- Applied NLP preprocessing — tokenization, stopword removal, lemmatization — improving model precision on real-world conversational inputs.

ML Pipeline – Classification & Prediction | Python · Scikit-learn · Pandas · Matplotlib · XGBoost

- Designed reusable end-to-end ML pipelines covering data ingestion, cleaning, feature engineering, and evaluation — reducing setup time by 50% for new datasets.
- Benchmarked Logistic Regression, Random Forest, and XGBoost using GridSearchCV; selected best model by F1-score and ROC-AUC. Visualized results with Matplotlib & Seaborn.

EDUCATION

Bachelor of Science in Computer Science | University of Education, Lahore, Pakistan | Sep 2022 – Oct 2026 (Expected)

Specializations: Artificial Intelligence · MERN Stack Web Development

CERTIFICATIONS

Certified Machine Learning – Coursera (2024) | Advanced Web Development – MERN Stack – Coursera (2024)

ADDITIONAL INFORMATION

Languages: English (Professional), Urdu (Native), Punjabi (Native) **Interests:** Deep learning research, NLP applications, AI-powered web systems, open-source ML